

Poultry ERP Developer Documentation

Complete Developer Documentation — Module Architecture & Specification

Prepared for: SRIRAM6369791334/software

August 2026

- 1. [Document Purpose](#)
- 2. [System Overview](#)
- 3. [Complete Module List](#)
 - 3.1 [Master Data Modules](#)
 - 3.2 [Farm Operations Module](#)
 - 3.3 [Warehouse / Inventory Module](#)
 - 3.4 [Procurement Module](#)
 - 3.5 [Sales & Billing Module](#)
 - 3.6 [Finance & Accounts Module](#)
 - 3.7 [Staff / HR Module \[NEW — full module\]](#)
 - 3.8 [Logistics Module](#)
 - 3.9 [Reports & Analytics Module](#)
 - 3.10 [Admin / System Module](#)
- 4. [Recommended Database Relationship Summary \(New Modules\)](#)
- 5. [Suggested Build Priority \(Phased Roadmap\)](#)
- 6. [Summary](#)

1. Document Purpose

This document defines the **complete module architecture** for a full-scale Poultry Business Management System — covering farm operations, warehouse/inventory, procurement, sales & billing, finance, staff/HR, logistics, retail shop POS, and system administration.

It is based on analysis of the existing codebase (Laravel 12, PHP 8.2, MySQL) and extends it with the modules required for a **complete, production-grade** poultry ERP. Existing modules are marked **[BUILT]**; new recommended modules are marked **[NEW]**; partially built modules are marked **[PARTIAL]**.

2. System Overview

Item	Detail
Framework	Laravel 12 (PHP 8.2)
Frontend	Blade + Tailwind CSS + Vite
Auth	Session-based (web) + Sanctum (API)
Authorization	spatie/laravel-permission (role + granular permission per action)
Database	MySQL (via Eloquent ORM)
PDF generation	barryvdh/laravel-dompdf
Queue	Laravel Queue (database/redis driver)

3. Complete Module List

3.1 Master Data Modules

3.1.1 Warehouse / Farm Master [BUILT]

Stores physical farm/warehouse locations used for stock and batch tracking.

Fields: `id, name, location, capacity, type (Farm/Warehouse/Cold-Storage), contact_person, phone, is_active`

Key relations: `hasMany(Batch)` , `hasMany(StockLedger)`

3.1.2 Shop Master [NEW]

Retail outlet master for direct-to-consumer sales points.

Fields: `id, shop_name, address, gst_number, phone, manager_staff_id, opening_date, is_active`

Key relations: `belongsTo(Staff, manager_staff_id)` , `hasMany(ShopSale)` , `hasMany(ShopStockTransfer)`

3.1.3 Vendor Master [BUILT]

Suppliers of birds, feed, medicine, and other inputs.

Fields: `firm_name, is_shop, gst_number, location, contact_person, phone, route, pending_amount`

3.1.4 Dealer Master [BUILT]

Bulk buyers / distributors who purchase birds for resale.

Fields: `firm_name, gst_number, location, contact_person, phone, route_id, pending_amount`

3.1.5 Customer Master [BUILT]

Retail/wholesale end customers.

Fields: `name, phone, address, gst_number, route_id, type, balance`

3.1.6 Route Master [BUILT]

Delivery route definitions tying dealers/customers to drivers and vehicles.

3.1.7 Item / Product Master [BUILT — needs extension]

Feed, medicine, chicks, eggs, live birds, by-products.

Extension needed: item category (Feed/Medicine/Live-Bird/Consumable), HSN code, GST slab, minimum stock, batch-linked or not.

3.1.8 Staff Master [NEW]

Employee master record — foundation for the HR module.

Fields: `id, employee_code, name, phone, address, designation, department, date_of_joining, salary_type (Fixed/Daily/Piece-rate), basic_salary, bank_account_no, ifsc_code, aadhaar_no, is_active, user_id (nullable FK to login account)`

3.2 Farm Operations Module

3.2.1 Batch / Flock Management [BUILT — 2 overlapping models: `Batch` , `BirdBatch`]

Tracks a flock from placement to closure.

Recommendation: consolidate `Batch` and `BirdBatch` into a single canonical model — currently duplicated, causing ambiguity between inventory batches and Day-Load batches.

Fields: `batch_code, placement_date, initial_count, current_count, breed, avg_placement_weight, status (Active/Closed), warehouse_id`

3.2.2 Daily Mortality Entry [BUILT]

Records bird deaths per batch per day, auto-decrements `current_count` .

3.2.3 Daily Consumption Entry [BUILT]

Feed / water / medicine consumption per batch per day; deducts from `StockLedger` .

3.2.4 Weight Tracking [PARTIAL]

`avg_weight` field exists but no dedicated time-series log.

Recommendation — new table `batch_weight_logs` : `batch_id, log_date, sample_size, avg_weight_g, recorded_by`

3.2.5 Vaccination & Medicine Schedule [NEW]

Critical for poultry farm compliance and bird health.

New table `vaccination_schedules` : `id`, `batch_id`, `vaccine_name`, `scheduled_day` (age in days), `scheduled_date`, `status` (Pending/Done/Missed), `administered_by`, `dose`, `remarks`

New table `vaccination_templates` : predefined age-based vaccination programs per breed, auto-applied when a batch is created.

3.2.6 FCR (Feed Conversion Ratio) Module [NEW]

Core poultry KPI = Total Feed Consumed (kg) ÷ Total Weight Gain (kg).

Computed report, not a new table — pulls from `Consumption` + `batch_weight_logs` + `Mortality`. Output: daily/weekly/batch-close FCR, benchmarked against breed standard.

3.2.7 Farm Environment Log [NEW — optional/advanced]

`farm_environment_logs` : `warehouse_id`, `log_date`, `log_time`, `temperature_c`, `humidity_pct`, `ammonia_ppm`, `recorded_by` — for climate-controlled broiler sheds.

3.3 Warehouse / Inventory Module

3.3.1 Multi-Warehouse Stock [BUILT]

3.3.2 Stock Ledger (in/out transactions) [BUILT]

3.3.3 Reorder Level Alert [PARTIAL] — `reorder_level` column exists; notification trigger not confirmed wired to `NotificationController`.

3.3.4 Stock Transfer Between Warehouses [NEW]

`stock_transfers` : `id`, `from_warehouse_id`, `to_warehouse_id`, `item_id`, `quantity`, `transfer_date`, `status` (Pending/In-Transit/Received), `initiated_by`, `received_by`

3.3.5 Batch-wise Stock Valuation [NEW]

Weighted-average cost valuation per item per warehouse, recalculated on every stock-in transaction. Feeds into P&L accuracy.

3.4 Procurement Module

3.4.1 Vendor Purchase Entry [BUILT]

3.4.2 Dealer Purchase Entry [BUILT]

3.4.3 Purchase Order → GRN Flow [NEW]

Formal procurement flow, currently missing — purchases go straight to invoice.

- `purchase_orders` : `id`, `vendor_id`, `po_number`, `po_date`, `expected_delivery_date`, `status` (Draft/Sent/Partially-Received/Closed)
- `purchase_order_items` : `po_id`, `item_id`, `ordered_qty`, `rate`
- `goods_receipt_notes` (GRN): `id`, `po_id`, `grn_number`, `received_date`, `received_by`
- `grn_items` : `grn_id`, `item_id`, `received_qty`, `rejected_qty`, `remarks`

3.4.4 Purchase Return / Rejection Tracking [NEW]

`purchase_returns` : `id`, `purchase_id/grn_id`, `item_id`, `quantity`, `reason`, `return_date`, `credit_note_no`

3.5 Sales & Billing Module

3.5.1 Day-Load Billing (Dealer↔Vendor real-time) [BUILT — strongest module]

3.5.2 Daily Billing (Customer) [BUILT]

3.5.3 Weekly Billing (Dealer, with GST) [BUILT]

3.5.4 Shop / Retail POS Billing [NEW]

Point-of-sale flow for walk-in shop customers, distinct from dealer/customer credit billing.

- `shop_sales` : `id`, `shop_id`, `sale_date`, `customer_name` (nullable, walk-in), `payment_mode` (Cash/UPI/Card), `total_amount`, `gst_amount`, `net_amount`

- `shop_sale_items`: `sale_id`, `item_id`, `quantity`, `rate`, `amount`
- Requires: barcode/quick-item lookup UI, daily shop cash reconciliation tying into Cash & Bank Ledger.

3.5.5 Credit / Cash Sale Distinction [PARTIAL]

`payment_mode` exists on bills; needs a unified credit-limit check per customer/dealer before allowing new credit sale.

3.5.6 Sales Return / Credit Note [NEW]

`sales_returns`: `id`, `bill_type` (Daily/Weekly/Shop), `bill_id`, `item_id`, `quantity`, `amount`, `reason`, `credit_note_no`, `return_date`

3.6 Finance & Accounts Module

3.6.1 Cash & Bank Ledger [BUILT]

3.6.2 Customer / Dealer / Vendor Payment Tracking [BUILT]

3.6.3 EMI Management [BUILT]

3.6.4 Expense Tracking [BUILT]

3.6.5 Profit / Loss Dashboard [BUILT]

3.6.6 Bank Reconciliation [NEW]

`bank_reconciliations`: `id`, `bank_statement_date`, `statement_reference`, `cash_bank_ledger_id`, `matched` (bool), `difference_amount`, `reconciled_by`, `reconciled_at` — imports bank statement CSV, auto-matches against ledger entries.

3.6.7 Ledger / Trial Balance / Balance Sheet Export [NEW]

Standard double-entry style summary reports generated from existing ledger tables — Trial Balance, P&L Statement, Balance Sheet — exportable as PDF/Excel for statutory use.

3.6.8 GST Filing-Ready Reports [NEW]

GSTR-1 (sales) and GSTR-2 (purchase) formatted export — extends existing GST% calculation into filing-compliant CSV/JSON.

3.7 Staff / HR Module [NEW — full module]

3.7.1 Staff Attendance

`staff_attendance`: `id`, `staff_id`, `date`, `status` (Present/Absent/Half-Day/Leave), `check_in`, `check_out`, `marked_by`

3.7.2 Salary / Payroll

`payroll_runs`: `id`, `month`, `year`, `status` (Draft/Processed/Paid) `payslips`: `payroll_run_id`, `staff_id`, `basic_salary`, `days_present`, `deductions`, `advances_deducted`, `net_pay`, `paid_at`

3.7.3 Staff Advance / Loan Tracking

`staff_advances`: `id`, `staff_id`, `amount`, `date`, `reason`, `status` (Outstanding/Recovered), `recovered_amount`

3.7.4 Role-Based Task Assignment

Extends existing spatie permission system with a lightweight `tasks` table: `id`, `assigned_to` (`staff_id`), `title`, `description`, `due_date`, `status`, `related_module`, `related_id`

3.8 Logistics Module

3.8.1 Vehicle Master [BUILT]

3.8.2 Driver Master [BUILT]

3.8.3 Route-wise Delivery Plan [PARTIAL]

3.8.4 Trip / Delivery Log [NEW]

delivery_trips : id, vehicle_id, driver_id, route_id, trip_date, start_odometer, end_odometer, start_time, end_time, status trip_stops : trip_id, dealer_id/customer_id, sequence_no, delivered_qty, delivery_status

3.8.5 Fuel / Maintenance Expense per Vehicle [NEW]

vehicle_expenses : id, vehicle_id, expense_type (Fuel/Service/Repair/Insurance), date, amount, odometer_reading, notes

3.9 Reports & Analytics Module

Report	Status
Inventory Analytics	[BUILT]
Profit Reports (weekly/monthly/batch)	[BUILT]
Dealer/Vendor Outstanding Report	[BUILT]
Batch-wise Profitability Report	[NEW] — links Batch → feed cost + medicine cost + mortality loss vs sale revenue
Vendor/Dealer Performance Comparison	[NEW] — rate trends, on-time payment score, purchase volume trends
FCR Trend Report	[NEW] — see 3.2.6
Shop-wise Sales Report	[NEW] — once Shop POS module exists
Staff Attendance & Payroll Summary	[NEW]

3.10 Admin / System Module

3.10.1 User Management, Roles, Permissions [BUILT — strong]

3.10.2 Activity Log [BUILT]

3.10.3 Notification System [BUILT]

3.10.4 Secure Backup & Audit Export [NEEDS REBUILD]

Current implementation (public/db_exporter.php , public/db_importer.php) is **unauthenticated and publicly reachable** — full database export/import with no login gate. This must be replaced with:

- An Artisan command (php artisan db:backup) run via CLI/cron only, OR
- A web route wrapped in auth + permission:manage backups + super-admin role + IP allowlist
- Removal of public/poultry_db (11).sql from the public web root entirely
- Removal or hardening of the live-deploy-sync-2026 route (currently runs live migrations + data mutation from a permission:create bills -gated GET request — should be a protected Artisan command instead)

4. Recommended Database Relationship Summary (New Modules)

Staff 1---N StaffAttendance
Staff 1---N StaffAdvance
Staff 1---N Payslip
PayrollRun 1---N Payslip
Shop 1---N ShopSale
ShopSale 1---N ShopSaleItem
PurchaseOrder 1---N PurchaseOrderItem
PurchaseOrder 1---N GoodsReceiptNote
GoodsReceiptNote 1---N GrnItem
Batch 1---N VaccinationSchedule
Batch 1---N BatchWeightLog
Warehouse 1---N FarmEnvironmentLog
Vehicle 1---N DeliveryTrip
DeliveryTrip 1---N TripStop
Vehicle 1---N VehicleExpense

5. Suggested Build Priority (Phased Roadmap)

Phase	Modules	Rationale
Phase 1 — Security Fix	Remove/secure db_exporter, db_importer, live-deploy-sync route	Live data exposure risk — must fix before any further deployment
Phase 2 — Farm Health	Vaccination Schedule, Weight Log, FCR Report	Core poultry-specific value, currently biggest functional gap
Phase 3 — Staff/HR	Staff Master, Attendance, Payroll, Advances	Needed for daily farm/shop labor operations
Phase 4 — Shop POS	Shop Master, Shop Sale, Shop Stock Transfer	Enables retail revenue channel
Phase 5 — Procurement Formalization	PO/GRN, Purchase Return	Tightens vendor purchase control
Phase 6 — Logistics	Delivery Trip, Vehicle Expense	Improves route/vehicle cost visibility
Phase 7 — Finance Depth	Bank Reconciliation, Trial Balance, GST filing export	Statutory/audit readiness
Phase 8 — Analytics	Batch Profitability, Vendor/Dealer Comparison, Shop Sales Report	Management decision support

6. Summary

The existing codebase already implements a mature **billing and financial ledger core** (Day-Load, Daily, Weekly billing, Cash-Bank Ledger, Payments, EMI) — this is the strongest part of the system.

To become a **complete poultry ERP**, the system needs four new pillars: **Farm Health (vaccination/FCR)**, **Staff/HR**, **Shop POS**, and **formal Procurement (PO/GRN)** — plus finance-depth features (bank reconciliation, statutory reports) and two urgent security fixes on the existing backup scripts and live-deploy route.